MIL-M-28787/160A(EC)
16 July 1982
SUPERSEDING
MIL-M-28787/160(NAVY)
14 November 1974

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

REGISTER, SHIFT, DIGITAL, KEY CODE EBH

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic shift register module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3 REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN ELEX 811, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 1.0 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be EBH.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/160-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2)

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1686. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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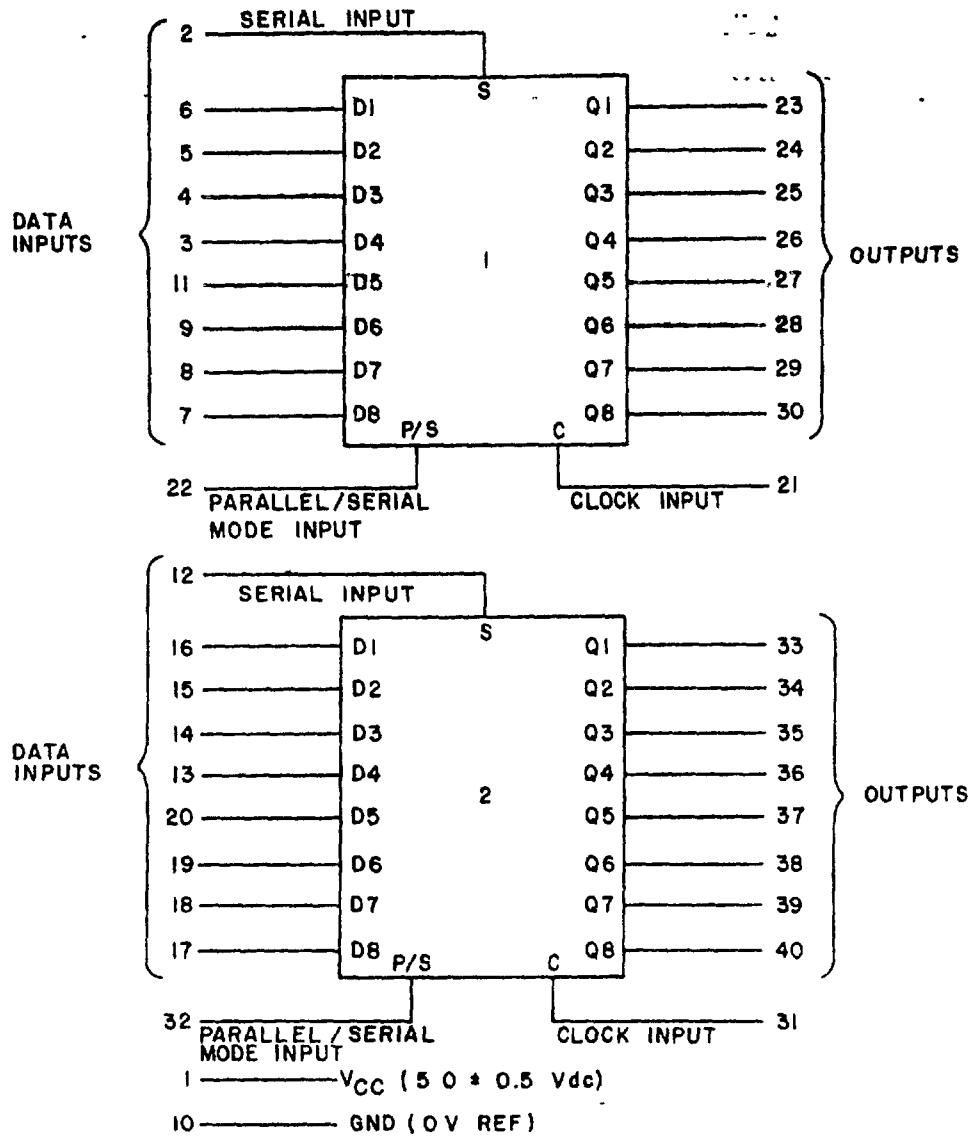


FIGURE 1. EBH module function diagram, shift register.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A16	V_{OUT} (low loaded)	II	S	0.0	0.6	0.0	0.6	0.0	0.6	0.0	0.6	V
B1-B18	Input breakdown current	III	S	0	100	0	100	0	100	0	100	µA
B19-B24	Input breakdown current	III	S	0	400	0	400	0	400	0	400	µA
C	Isolation	IV	S	10	∞	1	∞	10	∞	1	∞	MΩ
D1-D16	Output voltage, low level	V	X	0.0	0.3	0.0	0.3	0.0	0.3	0.0	0.3	V
D17-D32	Output voltage, high level	V	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
E1-E20	Functional shift right test	VI	X	← See tables VI and VII and figure 8 →								
F1-F18	Input current, low level	VIII	X	50	200	50	200	50	200	50	200	µA
F19-F24	Input current, low level	VIII	X	50	800	50	800	50	800	50	800	µA
G1-G18	Input current, high level	IX	X	0	10	0	10	0	10	0	10	µA
G19-G24	Input current, high level	IX	X	0	40	0	40	0	40	0	40	µA
H1-H16	Output short circuit current	X	X	3	15	3	15	3	15	3	15	mA
J1	Power supply current, low level	XI	S	2	36	2	36	2	36	2	36	mA
J2	Power supply current, high level	XI	S	2	36	2	36	2	36	2	36	mA
K1-K18	Transition time (t_{TLH}), C_L = 40 pF	XII	S	5	60	5	60	5	60	5	60	ns
K19-K36	Transition time (t_{TLH}), C_L = 140 pF	XII	Q	10	100	10	100	10	100	10	100	ns
L1-L18	Transition time (t_{THL}), C_L = 40 pF	XIII	S	2	30	2	30	2	30	2	30	ns
L19-L36	Transition time (t_{THL}), C_L = 140 pF	XIII	Q	5	60	5	60	5	60	5	60	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
M1-M32	Propagation delay time (t_{PLH}), C_L = 40 pF	XIV	S	10	165	10	165	10	200	10	200	ns
M33-M64	Propagation delay time (t_{PLH}), C_L = 140 pF	XIV	Q	10	195	10	195	10	230	10	230	ns
N1-N32	Propagation delay time (t_{PHL}), C_L = 40 pF	XV	S	10	165	10	165	10	200	10	200	ns
N33-N64	Propagation delay time (t_{PHL}), C_L = 140 pF	XV	Q	10	195	10	195	10	230	10	230	ns
P1-P34	Register operation	XVI	S	800	1200	800	1200	800	1200	800	1200	ns
Q1-Q16	Output crosstalk, low level	XVII	Q	0	400	0	400	0	400	0	400	mV
R1-R16	Output crosstalk, high level	XVII	Q	0	400	0	400	0	400	0	400	mV
S	Power supply transient amplitude	XVIII	Q	0	300	0	300	0	300	0	300	mV
T	Power supply filter capacitance	XIX	S	1.65	2.75	1.65	2.75	1.65	2.75	1.65	2.75	µF
U1-U18	Input capacitance	XX	Q	2	15	2	15	2	15	2	15	pF
U19-U22	Input capacitance	XX	Q	2	30	2	30	2	30	2	30	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through U shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through U in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through P, and T shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through T shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through T shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/160-1 supersedes part number M28787/160. The two parts are interchangeable.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation	190 mW (outputs not loaded)
Supply voltage	5 Vdc ±10 percent
Input voltage	
Low level	-0.5 V minimum, 0.7 V maximum
High level	2.0 V minimum, V_{CC} maximum
Data input current	
Low level (V_{IL} = 0.0 V)	-0.2 mA maximum
Low level (V_{IL} = 0.3 V)	-0.18 mA maximum
High level (V_{IH} = 2.4 V)	10 µA maximum
High level (V_{IH} = 5.5 V)	100 µA maximum

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P/S mode and clock input current	
Low level (V_{IL} = 0.0 V)	-0.8 mA maximum
Low level (V_{IL} = 0.3 V)	-0.72 mA maximum
High level (V_{IH} = 2.4 V)	40 µA maximum
High level (V_{IH} = 5.5 V)	400 µA maximum
Clock input	
Frequency	2.5 MHz maximum
Pulse width	200 ns minimum
Trigger edge	Negative
Data setup time	
Low level	120 ns minimum
High level	100 ns minimum
Data hold time	0 ns minimum
P/S mode control setup time	
Low level	225 ns minimum
High level	200 ns minimum
Output voltage (Q1-Q3 and Q5-Q8)	
Low level (I_{OL} = 2.0 mA)	0.3 V minimum
High level (I_{OH} = -100 µA)	2.4 V minimum
Output voltage (Q4)	
Low level (I_{OL} = 1.8 mA)	0.3 V maximum
High level (I_{OH} = -90 µA)	2.4 V minimum
Q1-Q3 and Q5-Q8 output drive maximum capability	
Low level (V_{OL} = 0.3 V)	2.0 mA
High level (V_{OH} = 2.4 V)	-100 µA
Q4 output drive maximum capability	
Low level (V_{OL} = 0.3 V)	1.8 mA
High level (V_{OH} = 2.4 V)	-90 µA
Propagation delay time (t_{PLH})	
40 pF load	10 ns minimum, 200 ns maximum
140 pF load	10 ns minimum, 230 ns maximum
Propagation delay time (t_{PHL})	
40 pF load	10 ns minimum, 200 ns maximum
140 pF load	10 ns minimum, 230 ns maximum

6.5 Module function. This module contains two independent 8-input registers. With a high level applied to the parallel/serial mode control input (P/S), internal gates are enabled which allow entry of data through parallel inputs (D1 through D8). This mode permits parallel loading of the register for parallel transfer to outputs (Q1 through Q8) when a clock signal (C) is received. With a low level applied to the parallel/serial mode control input (P/S), the paralleled input gates are inhibited and the right shift operation is performed on data at the serial data input (S) when a clock signal (C) is received. Shift left operation may be obtained by using the parallel mode with external connections.

6.5.1 Parallel outputs. Paralleling of outputs is not permissible.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

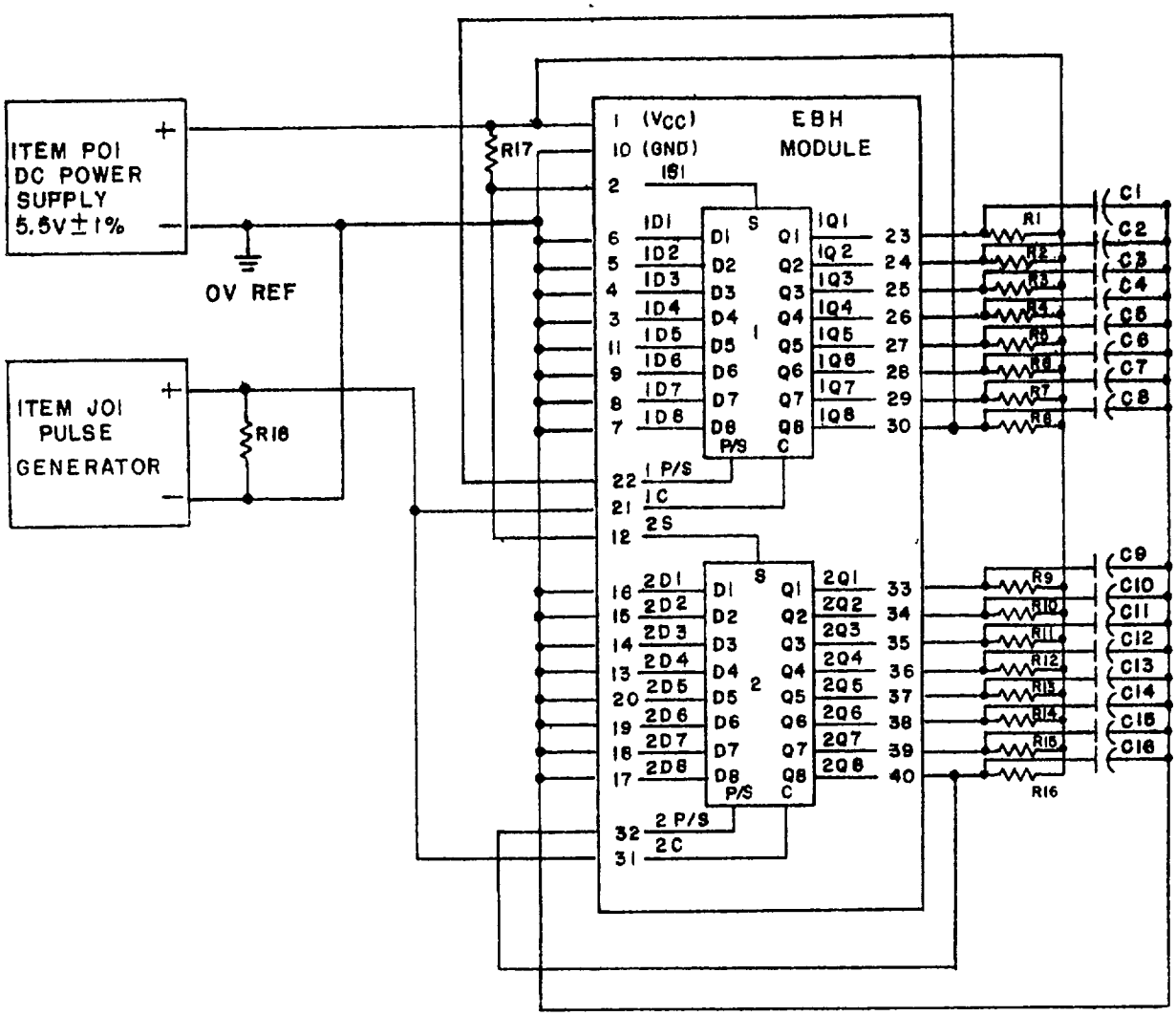
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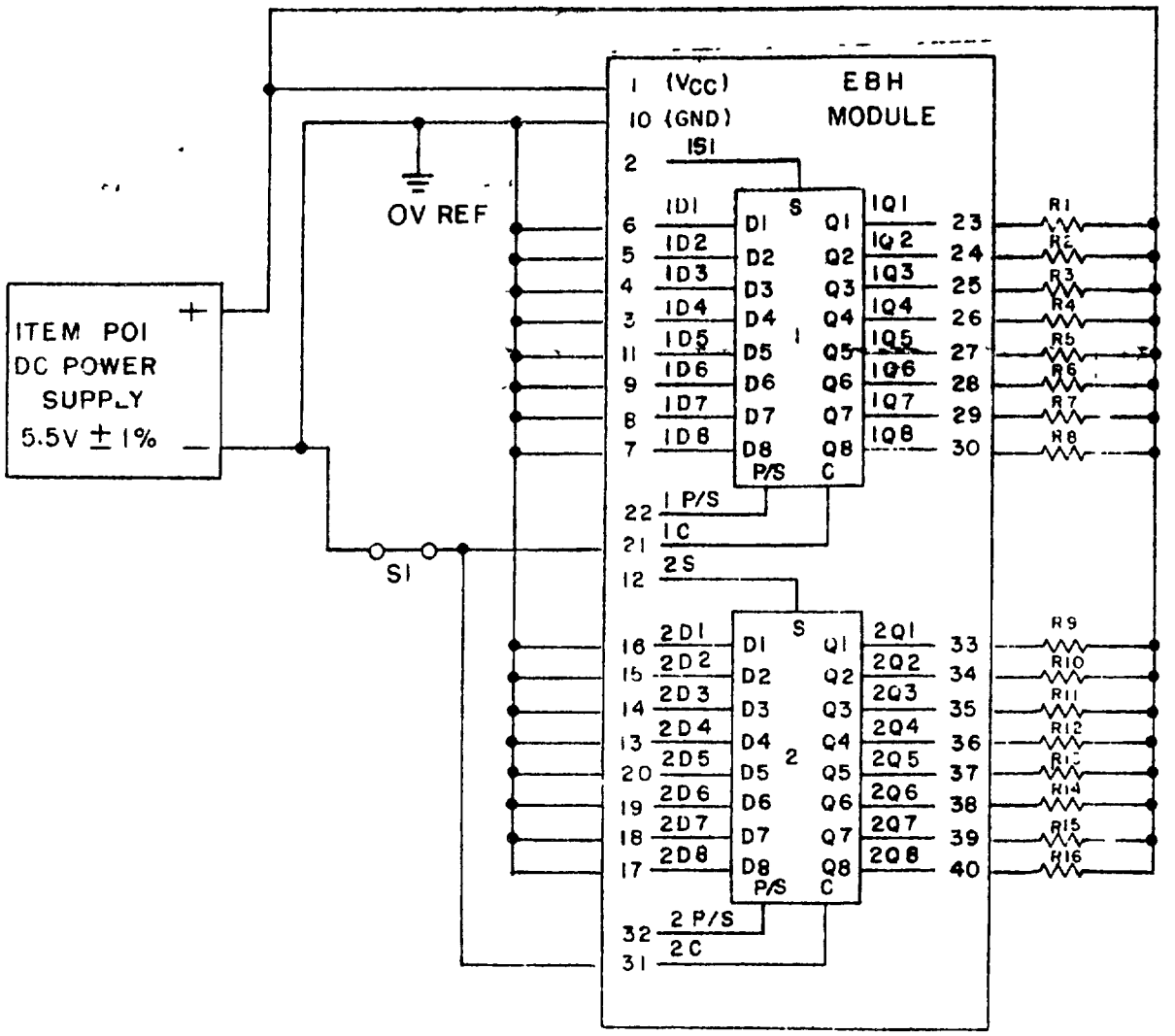
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NOTES.
1. R1-R17 = 2 75 kΩ ±1%, 1/8 W
2. C1-C16 = 100 pF ±20%
3. R18 = 51Ω ±5%, 1 W.
4. Pulse generator setup
   PA = 3 ±0.5 V
   PW = 500 ±50 ns
   $t_{TLH} = t_{THL}$ = 25 ±5 ns
   PRR = 1 0 ±0 1 MHz.

FIGURE 2. Operating temperature and life test circuit.

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NOTES
1. R1-R16 = 2 75 kΩ ±1%, 1/8 W.
2. S1 = SPST switch.
3. Open S1 for a minimum of 500 ns, then close it

FIGURE 3. Component temperature test circuit.

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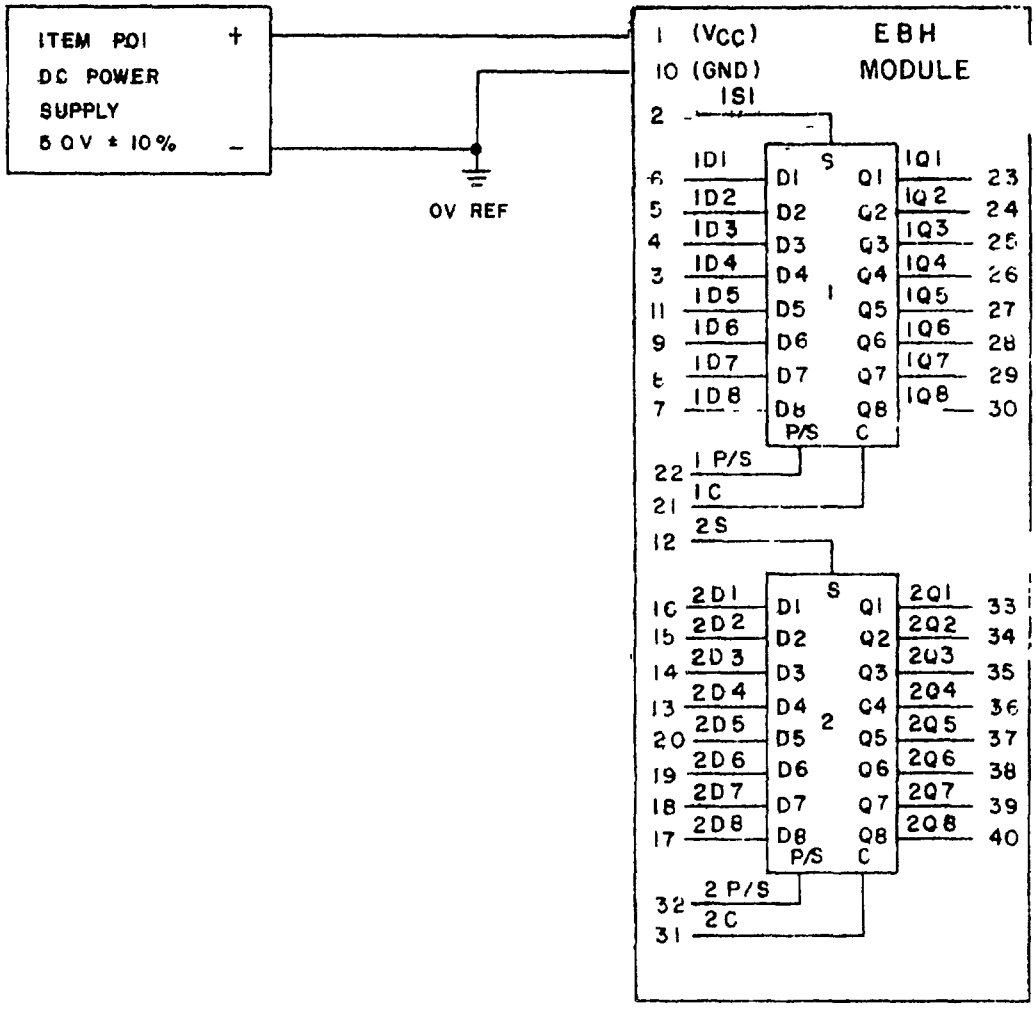


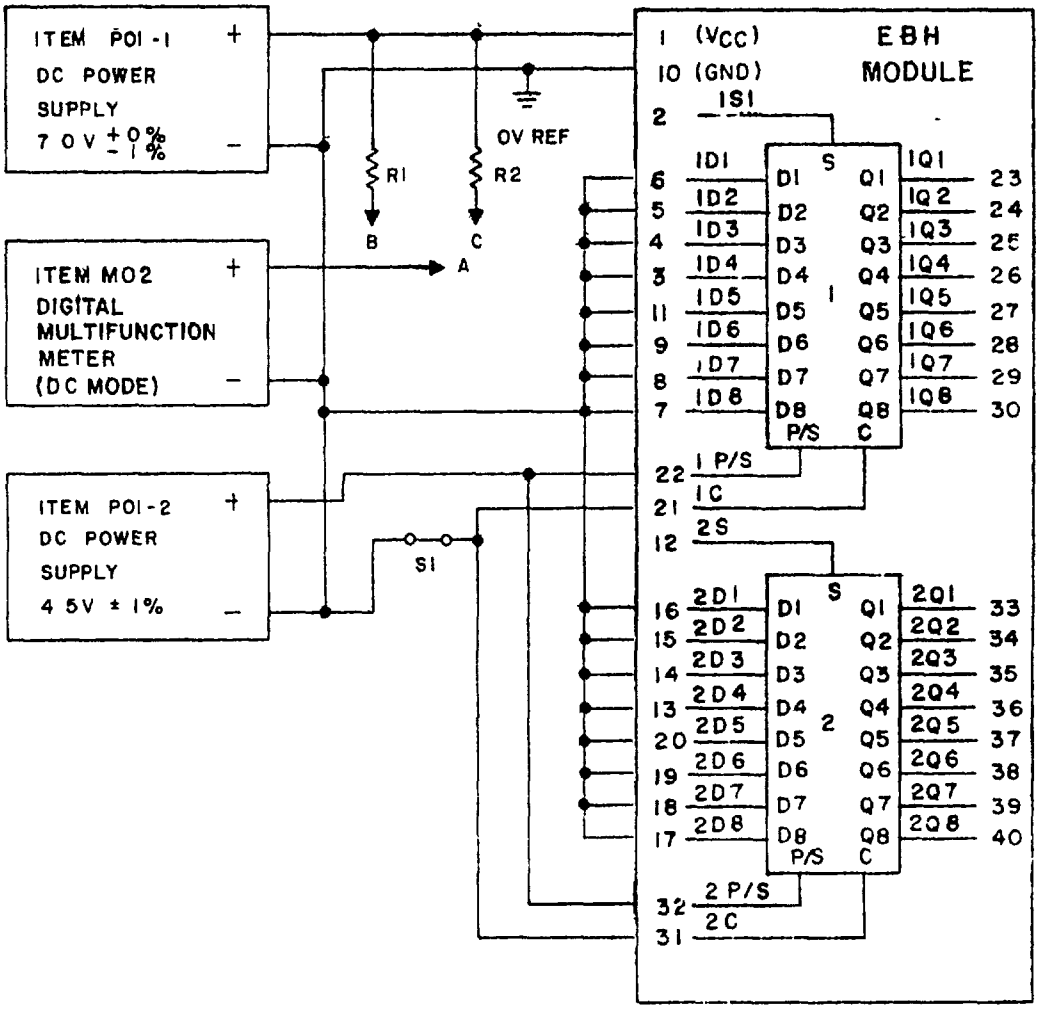
FIGURE 4. Vibration test circuit.

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TABLE II. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
A1	1Q1 V_{OUT} (low loaded)	23	23	Open	0.0	0.6	V	5
A2	1Q2	24	24	Open				
A3	1Q3	25	25	Open				
A4	1Q4	26	Open	26				
A5	1Q5	27	27	Open				
A6	1Q6	28	28					
A7	1Q7	29	29					
A8	1Q8	30	30					
A9	2Q1	33	33					
A10	2Q2	34	34					
A11	2Q3	35	35					
A12	2Q4	36	Open	36				
A13	2Q5	37	37	Open				
A14	2Q6	38	38					
A15	2Q7	39	39					
A16	2Q8	40	40					

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NOTES:
1. R1 = 2 75 kΩ ±1%, 1/8 W.
2. R2 = 3.09 kΩ ±1%, 1/8 W.
3. S1 = SPST switch.
4. Open S1 for a minimum of 500 ns, then close it and make measurements A1 through A16.
5. Measurements shall be recorded after point "B" or "C" has been connected to the appropriate module pin for at least 5 seconds

FIGURE 5 V_{OUT} (low-loaded) test circuit for table II.

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TABLE III. Input breakdown current test.

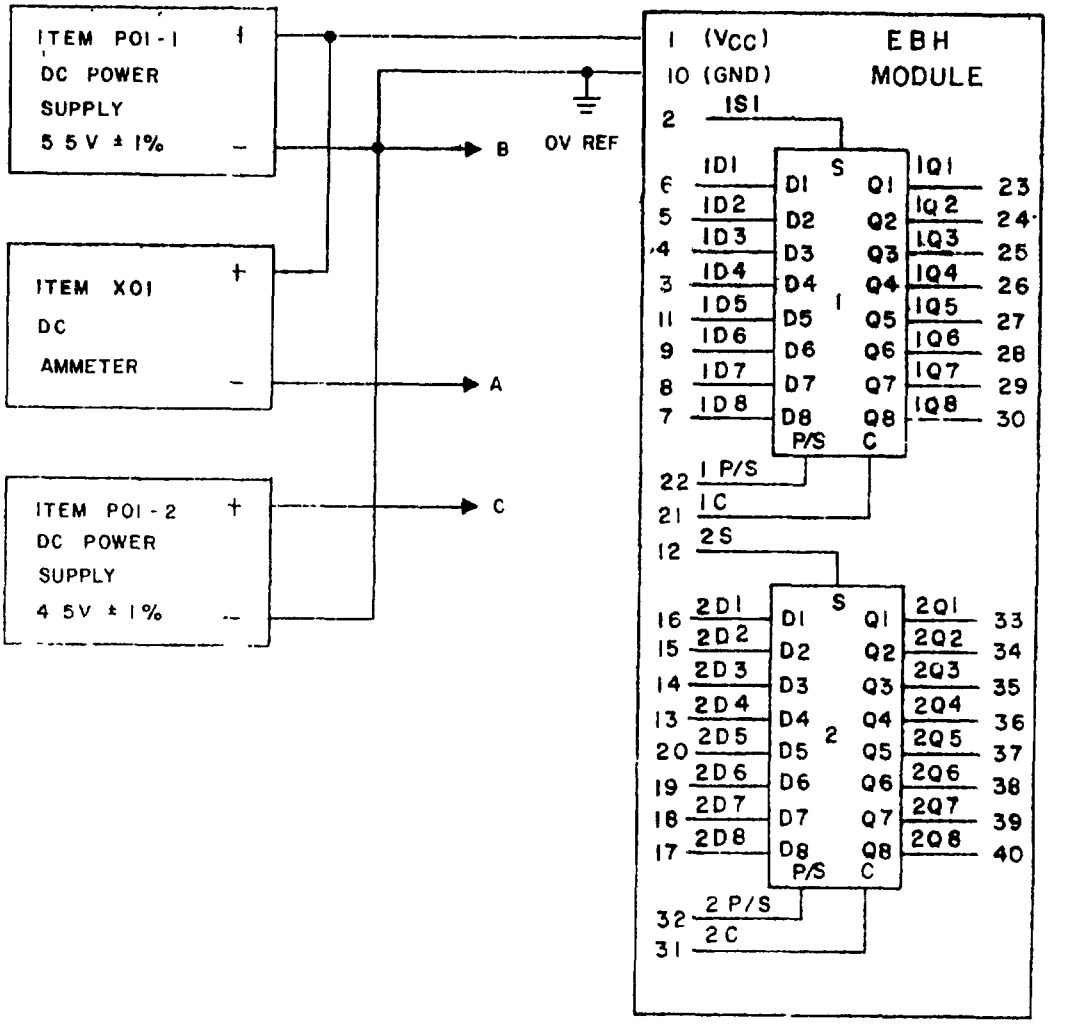
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
B1	1D1 Input breakdown current	6	22, 32	Open	0	100	µA	6
B2	1D2	5						
B3	1D3	4						
B4	1D4	3						
B5	1D5	11						
B6	1D6	9						
B7	1D7	8						
B8	1D8	7						
B9	2D1	16						
B10	2D2	15						
B11	2D3	14						
B12	2D4	13						
B13	2D5	20						
B14	2D6	19						
B15	2D7	18						
B16	2D8	17						
B17	1S	2	Open	22, 32				
B18	2S	12	Open	22, 32				
B19	1P/S	22	21	Open		400		

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TABLE III. Input breakdown current test - Continued.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
B20	2P/S	Input breakdown current	32	31	Open	0	400	µA	6
B21	1C	↓	21	Open	22	↓	↓	↓	↓
B22	1C		21	22	Open				
B23	2C		31	32	Open				
B24	2C		31	Open	32				

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

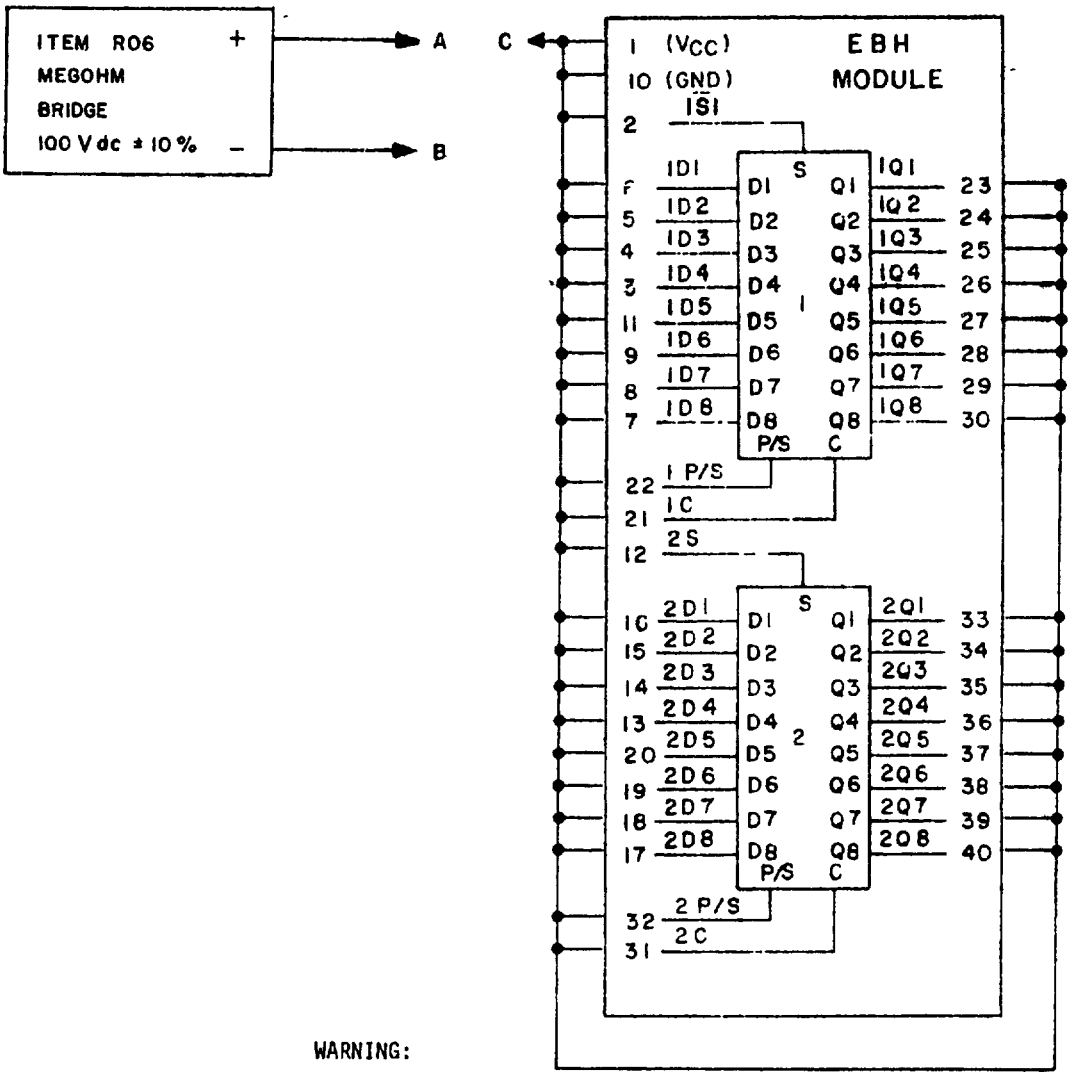
FIGURE 6. Input breakdown current test circuit for table III.

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TABLE IV. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C	Isolation, frame to module pins	C	Frame	10	∞	MΩ	7

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 7 Isolation test circuit for table IV

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TABLE V. Output voltage (low-level and high-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	1Q1 Output voltage, low level (parallel input, D input threshold low level, P/S threshold high level, threshold clock)	22	6,5,4,3,11, 9,8,7	23	0.0	0.3	V	8
D2	1Q2			24				
D3	1Q3			25				
D4	1Q4			26				
D5	1Q5			27				
D6	1Q6			28				
D7	1Q7			29				
D8	1Q8			30				
D9	2Q1	32	16,15,14,13, 20,19,18,17	33				
D10	2Q2			34				
D11	2Q3			35				
D12	2Q4			36				
D13	2Q5			37				
D14	2Q6			38				
D15	2Q7			39				
D16	2Q8			40				

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TABLE V. Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D17	1Q1 Output voltage, high level (parallel input, D input threshold high level, P/S threshold high level, threshold clock)	22,6,5,4,3, 11,9,8,7	16,15,14,13, 20,19,18,17	23	2.4	4.5	V	8
D18	1Q2			24				
D19	1Q3			25				
D20	1Q4			26				
D21	1Q5			27				
D22	1Q6			28				
D23	1Q7			29				
D24	1Q8			30				
D25	2Q1	32,16,15,14, 13,20,19,18, 17	Open	33				
D26	2Q2			34				
D27	2Q3			35				
D28	2Q4			36				
D29	2Q5			37				
D30	2Q6			38				
D31	2Q7			39				
D32	2Q8			40				

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TABLE VI. Functional shift right (low-level and high-level) test.

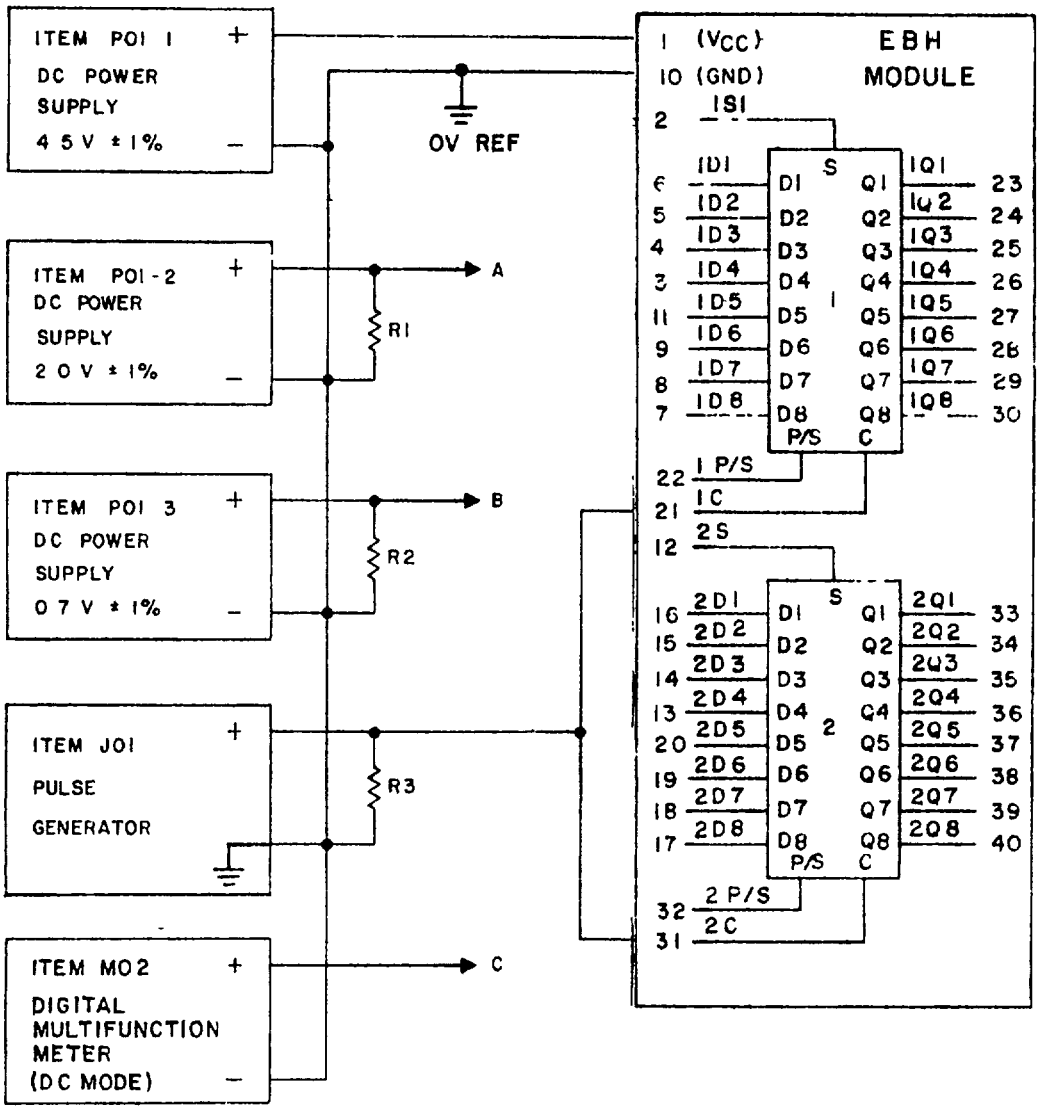
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	Functional shift right, low level (C, S, and P/S threshold conditions)	6,5,4,3,11,9,8,7,22,16,15,14,13,20,19,18,17,32	2, 12	See notes 2 and 6, figure 8		---	8
E2		6,16	2,5,4,3,11,9,8,7,22,12,15,14,13,20,19,18,17,32				
E3		2,5,12,15	6,4,3,11,9,8,7,22,16,14,13,20,19,18,17,32				
E4		2,4,12,14	6,5,3,11,9,8,7,22,16,15,13,20,19,18,17,32				
E5		2,3,12,13	6,5,4.11,9,8,7,22,16,15,14,20,19,18,17,32				
E6		2,11,12,20	6,5,4,3,9,8,7,22,16,15,14,13,19,18,17,32				
E7		2,9,12,19	6,5,4,3,11,8,7,22,16,15,14,13,20,18,17,32				
E8		2,8,12,18	6,5,4,3,11,9,7,22,16,15,14,13,20,19,17,32				
E9		2,7,12,17	6,5,4,3,11,9,8,22,16,15,14,13,20,19,18,32				
E10		2,12	6,5,4,3,11,9,8,7,22,16,15,14,13,20,19,18,17,32				

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TABLE VI. Functional shift right (low-level and high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E11	Functional shift right, high level (C, S, and P/S threshold conditions)	2,22,12,32	6,5,4,3,11,9, 8,7,16,15,14, 13,20,19,18,17	See notes 2 and 6, figure 8		---	8
E12		2,5,4,3,11,9,8,7, 12,15,14,13,20,19, 18,17	6,22,16,32				
E13		6,4,3,11,9,8,7, 16,14,13,20,19, 18,17	2,5,22,12,15, 32				
E14		6,5,3,11,9,8,7, 16,15,13,20,19, 18,17	2,4,22,12,14, 32				
E15		6,5,4,11,9,8,7, 16,15,14,20,19, 18,17	2,3,22,12,13, 32				
E16		6,5,4,3,9,8,7, 16,15,14,13,19, 18,17	2,11,22,12,20, 32				
E17		6,5,4,3,11,8,7, 16,15,14,13,20, 18,17	2,9,22,12,19, 32				
E18		6,5,4,3,11,9,7, 16,15,14,13,20, 19,17	2,8,22,12,18, 32				
E19		6,5,4,3,11,9,8, 16,15,14,13,20, 19,18	2,7,22,12,17, 32				
E20		6,5,4,3,11,9,8,7, 16,15,14,13,20, 19,18,17	2,22,12,32				

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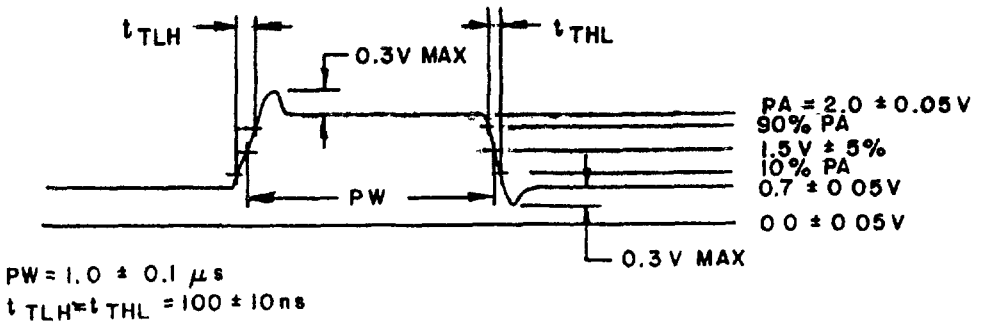


NOTES
1. R1 = R2 = R3 = 510 ±5%, 1 W
2 For test conditions E1-E20, the output logic levels for each circuit under test shall be detected simultaneously and shall be as required in table VII.
3 Load all outputs with loads as specified on figure 9

FIGURE 8 Output voltage (low-level and high-level) and functional shift right (low-level and high-level) test circuit for tables V and VI

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NOTES: (Continued)
4. Pulse generator setup (one shot mode).



5. Switching sequence for test conditions D1-D32.
   a. Apply one pulse to circuit by triggering item J01 and make measurements D1-D8.
   b. Repeat sequence for D9-D16.
   c Repeat sequence for D17-D24.
   d. Repeat sequence for D25-D32.
6. Switching sequence for test conditions E1-E20.
   a Test conditions E1-E10 and E11-E20 shall be performed in sequence.
   b. Make all module pin connections for each test condition per table VI and, using the item J01 in the single pulse mode, apply a single pulse before making each measurement
   c After each pulse, the voltage level of the outputs Q1-1 through Q8-2 shall correspond to the output state as specified in table VII. Verify that the low level (L) is less than 0.3 V and the high level (H) is greater than 2.4 V.
   d. Repeat sequence for test conditions E11-E20.

FIGURE 8 Output voltage (low-level and high-level) and functional shift right (low-level and high-level) test circuit for tables V and VI - Continued.

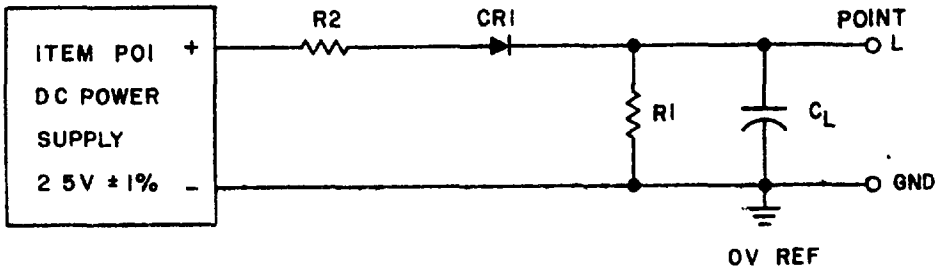
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TABLE VII. Functional shift right (low-level and high-level) output logic states.

Test condition	Clock input	Q1-1	Q2-1	Q3-1	Q4-1	Q5-1	Q6-1	Q7-1	Q8-1	Q1-2	Q2-2	Q3-2	Q4-2	Q5-2	Q6-2	Q7-2	Q8-2
E1	Pulse 1	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H
E2	Pulse 2	L	H	H	↓	⋮	⋮	⋮	⋮	L	H	H	⋮	⋮	⋮	⋮	⋮
E3	Pulse 3	H	L	H	↓	⋮	⋮	⋮	⋮	H	L	H	↓	⋮	⋮	⋮	⋮
E4	Pulse 4	⋮	H	L	↓	↓	⋮	⋮	⋮	⋮	H	L	⋮	⋮	⋮	⋮	⋮
E5	Pulse 5	⋮	⋮	H	L	↓	⋮	⋮	⋮	⋮	⋮	H	L	↓	⋮	⋮	⋮
E6	Pulse 6	⋮	⋮	⋮	H	L	↓	⋮	⋮	⋮	⋮	⋮	H	L	↓	⋮	⋮
E7	Pulse 7	⋮	⋮	⋮	⋮	H	L	↓	⋮	⋮	⋮	⋮	⋮	H	L	↓	⋮
E8	Pulse 8	⋮	⋮	⋮	⋮	⋮	H	L	↓	⋮	⋮	⋮	⋮	⋮	H	L	↓
E9	Pulse 9	⋮	⋮	⋮	⋮	⋮	H	H	L	⋮	⋮	⋮	⋮	⋮	H	H	L
E10	Pulse 10	↓	↓	↓	↓	↓	H	H	H	↓	↓	↓	↓	↓	H	H	H
E11	Pulse 1	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L
E12	Pulse 2	H	L	L	⋮	⋮	⋮	⋮	⋮	H	L	L	↓	⋮	⋮	⋮	⋮
E13	Pulse 3	L	H	L	↓	⋮	⋮	⋮	⋮	L	H	L	⋮	⋮	⋮	⋮	⋮
E14	Pulse 4	⋮	L	H	↓	↓	⋮	⋮	⋮	⋮	L	H	↓	⋮	⋮	⋮	⋮
E15	Pulse 5	⋮	⋮	L	H	↓	⋮	⋮	⋮	⋮	⋮	L	H	↓	⋮	⋮	⋮
E16	Pulse 6	⋮	⋮	⋮	L	H	↓	⋮	⋮	⋮	⋮	⋮	L	H	↓	⋮	⋮
E17	Pulse 7	⋮	⋮	⋮	⋮	L	H	↓	⋮	⋮	⋮	⋮	⋮	L	H	↓	⋮
E18	Pulse 8	⋮	⋮	⋮	⋮	⋮	L	H	↓	⋮	⋮	⋮	⋮	⋮	L	H	↓
E19	Pulse 9	⋮	⋮	⋮	⋮	⋮	L	L	H	⋮	⋮	⋮	⋮	⋮	L	L	H
E20	Pulse 10	↓	↓	↓	↓	↓	L	L	L	↓	↓	↓	↓	↓	L	L	L

1/ Q1-1 through Q8-2 output logic states are measured at output pins 23, 24, 25, 26, 27, 28, 29, 30, 33, 34, 35, 36, 37, 38, 39, and 40, respectively.

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NOTES
1. CR1 = 1N916 or equivalent.
2. R1 = 23.7 kΩ ±1%, 1/8 W for a current flow of 100 µA into point L when L is held at 2 4 V for module outputs Q1-Q3 and Q5-Q8.
   = 26.7 kΩ ±1%, 1/8 W for a current flow 90 µA into point L when L is held at 2.4 V for module output Q4.
3. R2 = Select value for current flow of 2.0 mA ±5% when point L is held at 0.3 V for module outputs Q1-Q3 and Q5-Q8.
   = Select value for current flow of 1.8 mA ±5% when point L is held at 0.3 V for module output Q4.
4. C_L = 40 ±5 pF (including probe capacitance) for test conditions K1-K18, L1-L18, M1-M32 and N1-N32.
   = 140 ±5 pF (including probe capacitance) for test conditions K19-K36, L19-L36, M33-M64 and N33-N64.
   = 50 ±5 pF (including probe capacitance) for test conditions Q1-Q16 and R1-R16
5. C_L is not required for dc electrical tests
6. The maximum lead length from point "L" to the appropriate module pin shall be 3 inches (76.2 mm).
7. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV

FIGURE 9. EBH logic load circuit

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TABLE VIII. Input current (low-level) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F1	1D1 Input current, low level	Open	6	Open	22	50	200	µA	10
F2	1D2		5						
F3	1D3		4						
F4	1D4		3						
F5	1D5		11						
F6	1D6		9						
F7	1D7		8						
F8	1D8		7						
F9	2D1		16		32				
F10	2D2		15						
F11	2D3		14						
F12	2D4		13						
F13	2D5		20						
F14	2D6		19						

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TABLE VIII. Input current (low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F15	2D7 Input current, low level	Open	18	Open	32	50	200	mA	10
F16	2D8		17	Open	32				
F17	1S		2	22	Open				
F18	2S		12	32	Open				
F19	1C		21	22	Open		800		
F20	1C		21	Open	22				
F21	2C		31	32	Open				
F22	2C		31	Open	32				
F23	1P/S		22	Open	21				
F24	2P/S		32	Open	31				

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TABLE IX. Input current (high-level) test.

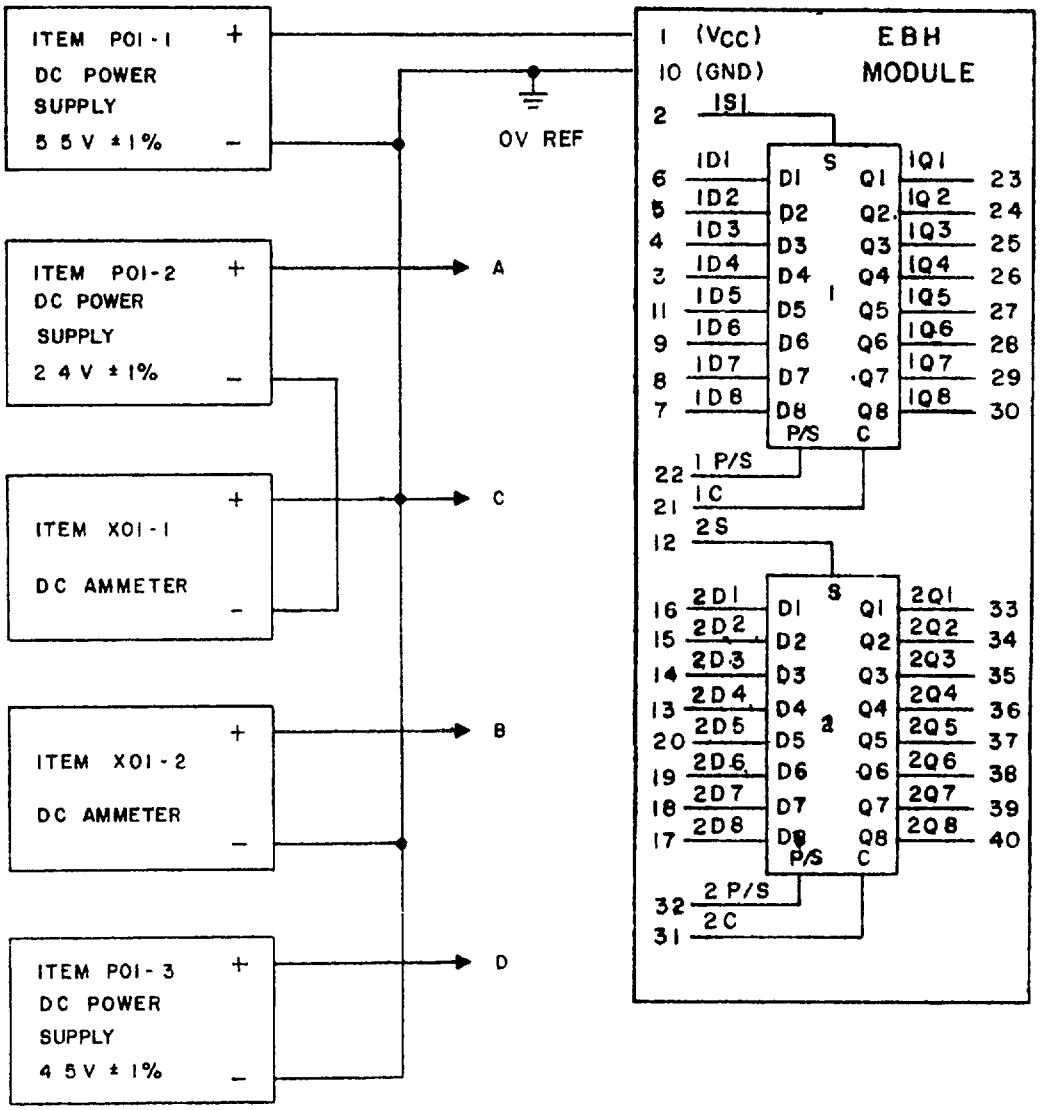
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	1D1 Input current, high level	6	Open	22	Open	0	10	µA	10
G2	1D2	5							
G3	1D3	4							
G4	1D4	3							
G5	1D5	11							
G6	1D6	9							
G7	1D7	8							
G8	1D8	7							
G9	2D1	16		32					
G10	2D2	15							
G11	2D3	14							
G12	2D4	13							
G13	2D5	20							
G14	2D6	19							
G15	2D7	18							

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TABLE IX. Input current (high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G16	2D8 Input current, high level	17	Open	32	Open	0	10	µA	10
G17	1S	2		Open	22				
G18	2S	12		Open	32				
G19	1C	21		Open	22		40		
G20	1C	21		22	Open				
G21	2C	31		Open	32				
G22	2C	31		32	Open				
G23	1P/S	22		21	Open				
G24	2P/S	32		31	Open				

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CAUTION:
Do not isolate the chassis of ammeter XO1 from earth ground.

NOTES:
1. Since items XO1-1 and XO1-2 are not used simultaneously, only one item, XO1, is required
2. Since the items PO1-2 and PO1-3 are not used simultaneously, only one item, PO1, is required
3. With point "A" disconnected, item XO1-1 should indicate zero current on the 1 µA range

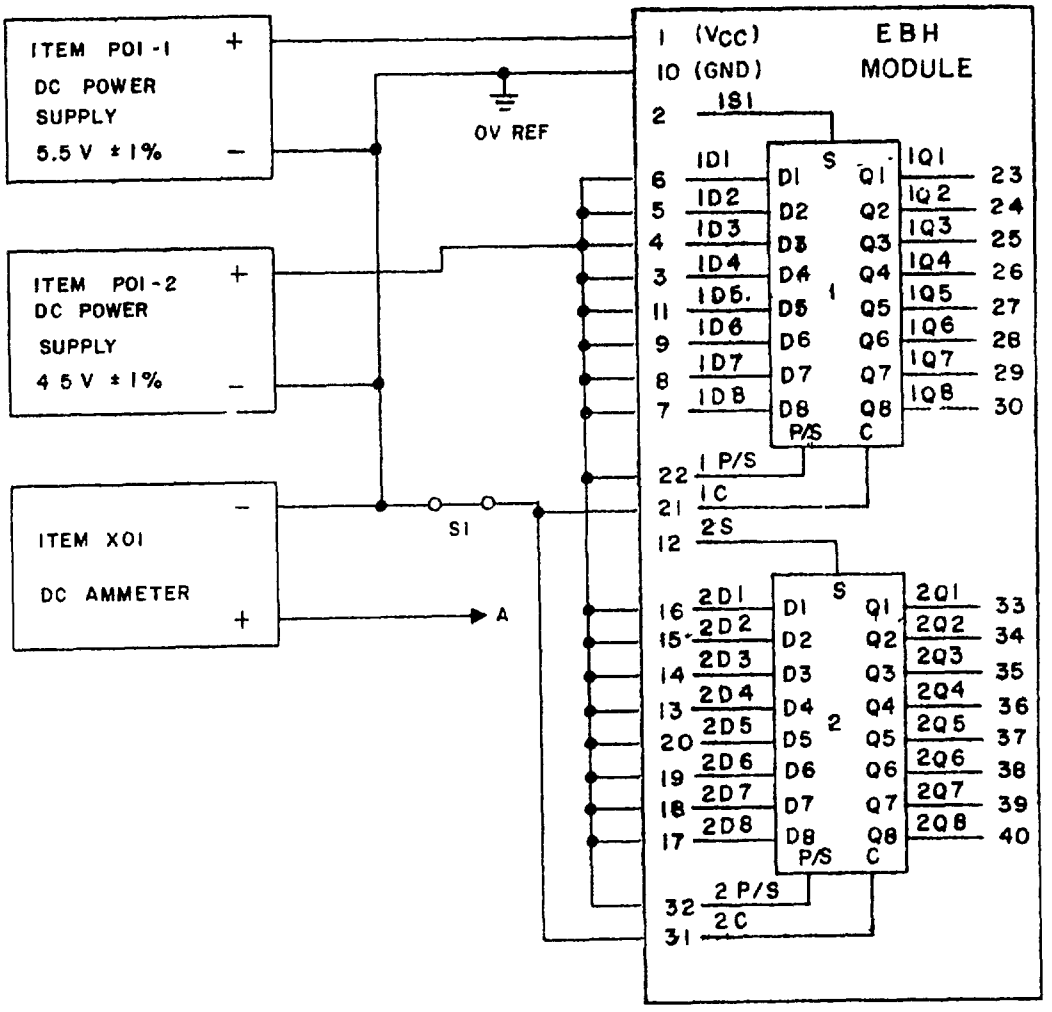
FIGURE 10. Input current (low-level and high-level) test circuit for tables VIII and IX.

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TABLE X. Output short circuit current test.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
H1	1Q1	Output short circuit current	23	3	15	mA	11
H2	1Q2		24				
H3	1Q3		25				
H4	1Q4		26				
H5	1Q5		27				
H6	1Q6		28				
H7	1Q7		29				
H8	1Q8		30				
H9	2Q1		33				
H10	2Q2		34				
H11	2Q3		35				
H12	2Q4		36				
H13	2Q5		37				
H14	2Q6		38				
H15	2Q7		39				
H16	2Q8		40				

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTES:
1. S1 = SPST switch.
2. With point "A" disconnected, item XO1 should indicate zero current on the 1 µA range.
3. When performing output short circuit tests, only one output shall be shorted at a time and the measurement taken within 5 seconds.
4. Open S1 for a minimum of 500 ns, then close it and make measurements H1 through H16

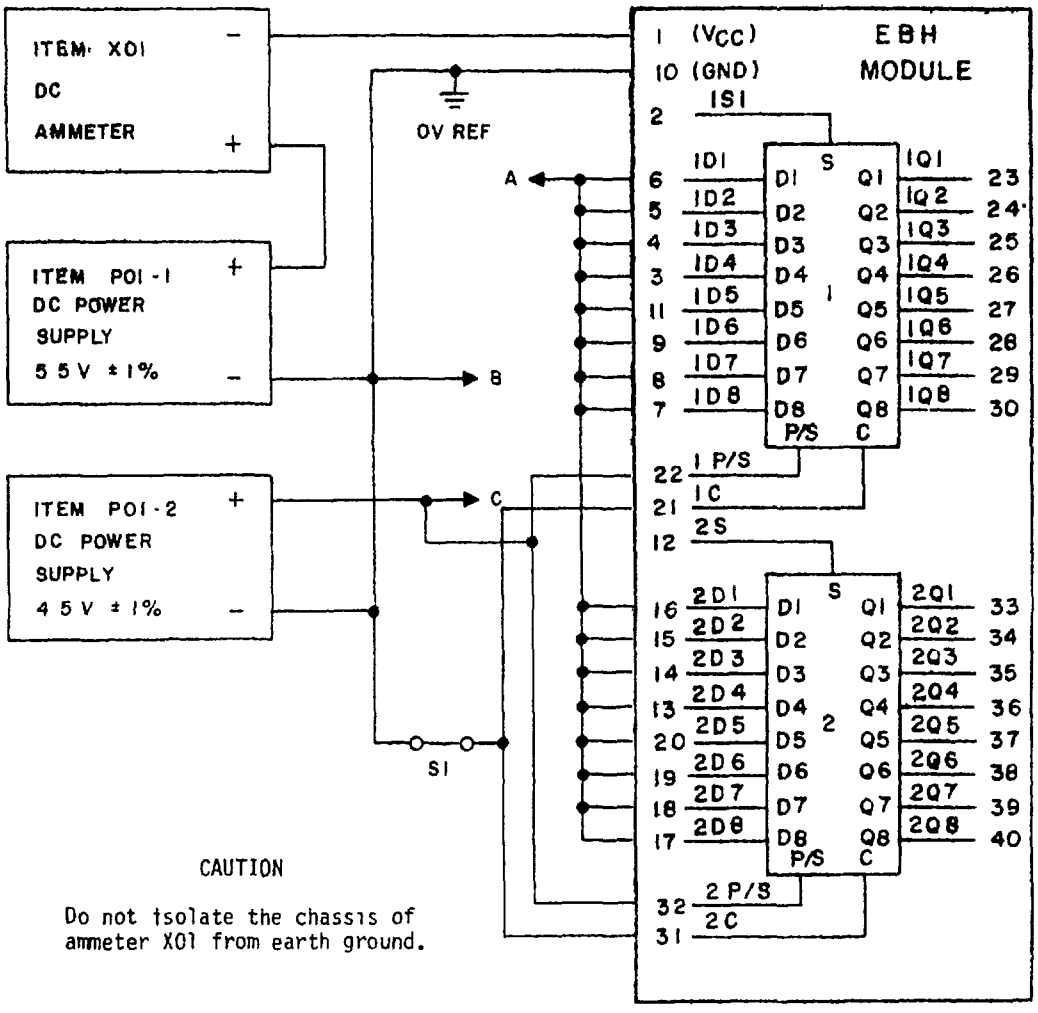
FIGURE 11. Output short circuit current test circuit for table X.

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TABLE XI. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	Power supply current, low level	B	A	Open	2	36	mA	12
J2	Power supply current, high level	C	Open	A	2	36	mA	12

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NOTES:
1 S1 = SPST switch.
2 For each measurement, connect circuit, open S1 for a minimum of 500 ns, then close it and take reading

FIGURE 12 Power supply current (low-level and high-level) test circuit for table XI

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TABLE XII. Transition time (t_{TLH}) test.

on		Parameter	Connect					Initial limits 25°C±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
	1Q1	Transition time (t_{TLH}), C_L = 40 pF	2	22	6,5,4,3,11, 9,8,7	21	23	5	60	ns	13,14
	1Q2						24				
	1Q3						25				
	1Q4						26				
	1Q5						27				
	1Q6						28				
	1Q7						29				
	1Q8						30				
	2Q1		12	32	16,15,14,13, 20,19,18,17	31	33				
	2Q2						34				
	2Q3						35				
	2Q4						36				
	2Q5						37				
	2Q6						38				
	2Q7						39				
	2Q8						40				
	1Q1		22,6,5,4,3,11, 9,8,7	Open	2	21	23				
	2Q1		32,16,15,14,13, 20,19,18,17	Open	12	31	33				

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TABLE XII. Transition time (t_{TLH}) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Initial limits 25°C±5°C ambient Min	Initial limits 25°C±5°C ambient Max	Unit	Test figure
K19	1Q1 Transition time (t_{TLH}), C_L = 140 pF	2	22	6,5,4,3,11, 9,8,7	21	23	10	100	ns	13,14
K20	1Q2					24				
K21	1Q3					25				
K22	1Q4					26				
K23	1Q5					27				
K24	1Q6					28				
K25	1Q7					29				
K26	1Q8					30				
K27	2Q1	12	32	16,15,14,13, 20,19,18,17	31	33				
K28	2Q2					34				
K29	2Q3					35				
K30	2Q4					36				
K31	2Q5					37				
K32	2Q6					38				
K33	2Q7					39				
K34	2Q8					40				
K35	1Q1	22,6,5,4,3,11, 9,8,7	Open	2	21	23				
K36	2Q1	32,16,15,14,13, 20,19,18,17	Open	12	31	33				

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TABLE XIII. Transition time (t_{THL}) test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
L1	1Q1 Transition time (t_{THL}), C_L = 40 pF	2	22	6,5,4,3, 11,9,8,7	21	23	2	30	ns	13,14,
L2	1Q2					24				
L3	1Q3					25				
L4	1Q4					26				
L5	1Q5					27				
L6	1Q6					28				
L7	1Q7					29				
L8	1Q8					30				
L9	2Q1	12	32	16,15, 14,13, 20,19, 18,17	31	33				
L10	2Q2					34				
L11	2Q3					35				
L12	2Q4					36				
L13	2Q5					37				
L14	2Q6					38				
L15	2Q7					39				

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TABLE XIII. Transition time (t_{THL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
L16	2Q8 Transition time (t_{THL}), C_L = 40 pF	12	32	16,15, 14,13, 20,19, 18,17	31	40	2	30	ns	13,14,
L17	1Q1 Transition time (t_{THL}), C_L = 40 pF	22,6,5, 4,3,11, 9,8,7	Open	2	21	23				
L18	2Q1 Transition time (t_{THL}), C_L = 40 pF	32,16, 15,14, 13,20, 19,18, 17	Open	12	31	33				
L19	2Q1 Transition time (t_{THL}), C_L = 140 pF	2	22	6,5,4,3, 11,9,8,7	21	23	5	60		
L20	1Q2					24				
L21	1Q3					25				
L22	1Q4					26				
L23	1Q5					27				
L24	1Q6					28				
L25	1Q7					29				
L26	1Q8					30				

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TABLE XIII. Transition time (t_{THL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
L27	2Q1 Transition time (t_{THL}), C_L = 140 pF	12	32	16,15, 14,13, 20,19, 18,17	31	33	5	60	ns	13,14
L28	2Q2					34				
L29	2Q3					35				
L30	2Q4					36				
L31	2Q5					37				
L32	2Q6					38				
L33	2Q7					39				
L34	2Q8					40				
L35	1Q1	22,6,5, 4,3,11, 9,8,7	Open	2	21	23				
L36	2Q1	32,16, 15,14, 13,20, 19,18,17	Open	12	31	33				

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TABLE XIV. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect: A to pin	B to pin	C to pin	D to pin	E to pin	Initial limits 25°±5°C ambient: Min	Max	Unit	Test figure
M1	1D1 Propagation delay time (t_{PLH}), C_L = 40 pF (minimum data hold time check at minimum clock time)	2	22	6,5,4,3, 11,9,8,7	21	23	10	165	ns	13,14
M2	1D2					24				
M3	1D3					25				
M4	1D4					26				
M5	1D5					27				
M6	1D6					28				
M7	1D7					29				
M8	1D8					30				
M9	2D1	12	32	16,15, 14,13, 20,19, 18,17	31	33				
M10	2D2					34				
M11	2D3					35				
M12	2D4					36				
M13	2D5					37				
M14	2D6					38				
M15	2D7					39				
M16	2D8					40				

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TABLE XIV. Propagation delay time (t_{PLH}) test - Continued.

Test condition		Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
M17	15	Propagation delay time (t_{PLH}), C_L = 40 pF (minimum data hold time check at minimum clock time)	22,6,5, 4,3,11, 9,8,7	Open	2	21	23	10	165	ns	13,14
M18	15						24				
M19	15						25				
M20	15						26				
M21	15						27				
M22	15						28				
M23	15						29				
M24	15						30				
M25	25		32,16, 15,14, 13,20, 19,18,17		12	31	33				
M26	25						34				
M27	25						35				
M28	25						36				
M29	25						37				
M30	25						38				
M31	25						39				
M32	25						40				

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TABLE XIV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
M33	1D1 Propagation delay time (t_{PLH}), C_L = 140 pF (minimum data hold time check at minimum clock time)	2	22	6,5,4,3, 11,9,8,7	21	23	10	195	ns	13,14
M34	1D2					24				
M35	1D3					25				
M36	1D4					26				
M37	1D5					27				
M38	1D6					28				
M39	1D7					29				
M40	1D8					30				
M41	2D1	12	32	16,15, 14,13, 20,19, 18,17	31	33				
M42	2D2					34				
M43	2D3					35				
M44	2D4					36				
M45	2D5					37				
M46	2D6					38				
M47	2D7					39				
M48	2D8					40				

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TABLE XIV. Propagation delay time (t_{PLH}) test - Continued.

Test condition		Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
M49	1S	Propagation delay time (t_{PLH}), C_L = 140 pF (minimum data hold time check at minimum clock time)	22,6,5, 4,3,11, 9,8,7	Open	2	21	23	10	195	ns	13,14
M50	1S						24				
M51	1S						25				
M52	1S						26				
M53	1S						27				
M54	1S						28				
M55	1S						29				
M56	1S						30				
M57	2S		32,16, 15,14, 13,20, 19,18,17		12	31	33				
M58	2S						34				
M59	2S						35				
M60	2S						36				
M61	2S						37				
M62	2S						38				
M63	2S						39				
M64							40				

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TABLE XV. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Connect: E to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
N1	1D1 Propagation delay time (t_{PHL}), C_L = 40 pF (minimum data hold time check at minimum clock time)	2	22	6,5,4,3, 11,9,8,7	21	23	10	165	ns	13,14
N2	1D2					24				
N3	1D3					25				
N4	1D4					26				
N5	1D5					27				
N6	1D6					28				
N7	1D7					29				
N8	1D8					30				
N9	2D1	12	32	16,15, 14,13, 20,19, 18,17	31	33				
N10	2D2					34				
N11	2D3					35				
N12	2D4					36				
N13	2D5					37				
N14	2D6					38				
N15	2D7					39				
N16	2D8					40				

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
N17	15 Propagation delay time (t_{PHL}), C_L = 40 pF (minimum data hold time check at minimum clock time)	22,6,5, 4,3,11, 9,8,7	Open	2	21	23	10	165	ns	13,14
N18	1S					24				
N19	1S					25				
N20	1S					26				
N21	1S					27				
N22	1S					28				
N23	1S					29				
N24	1S					30				
N25	2S	32,16, 15,14, 13,20, 19,18,17		12	31	33				
N26	2S					34				
N27	2S					35				
N28	2S					36				
N29	2S					37				
N30	2S					38				
N31	2S					39				
N32	2S					40				

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
N33	1D1 Propagation delay time (t_{PHL}), C_L = 140 pF (minimum data hold time check at minimum clock time)	2	22	6,5,4,3, 11,9,8,7	21	23	10	195	ns	13,14
N34	1D2					24				
N35	1D3					25				
N36	1D4					26				
N37	1D5					27				
N38	1D6					28				
N39	1D7					29				
N40	1D8					30				
N41	2D1	12	32	16,15, 14,13, 20,19, 18,17	31	33				
N42	2D2					34				
N43	2D3					35				
N44	2D4					36				
N45	2D5					37				
N46	2D6					38				
N47	2D7					39				
N48	2D8					40				

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

Test condition		Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
N49	11S	Propagation delay time (t_{PHL}), C_L = 140 pF (minimum data hold time check at minimum clock time)	22,6,5, 14,3,11, 9,8,7	Open	2	21	23	10	195	ns	13,14
N50	11S						24				
N51	11S						25				
N52	11S						26				
N53	11S						27				
N54	11S						28				
N55	11S						29				
N56	11S						30				
N57	12S		32,16, 15,14, 13,20, 19,18,17		12	31	33				
N58	12S						34				
N59	12S						35				
N60	12S						36				
N61	12S						37				
N62	12S						38				
N63	12S						39				
N64	12S						40				

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TABLE XVI. Register operation (input minimum setup time check) test.

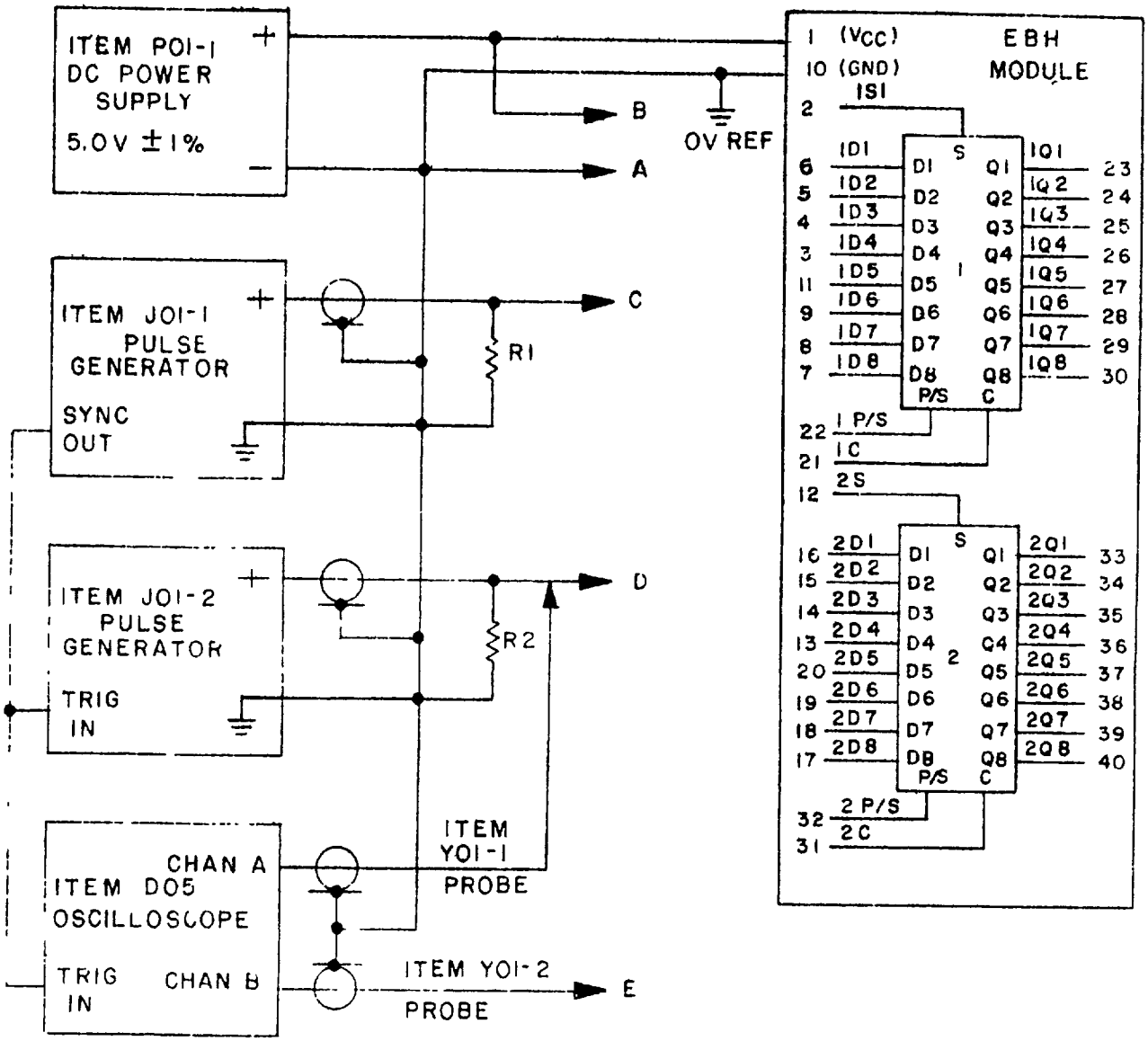
Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
P1	1D1 Register operation (input minimum setup time check)	2	22	6,5,4,3, 11,9,8,7	21	23	800	1200	ns	13,15
P2	1D2					24				
P3	1D3					25				
P4	1D4					26				
P5	1D5					27				
P6	1D6					28				
P7	1D7					29				
P8	1D8					30				
P9	2D1	12	32	16,15, 14,13, 20,19, 18,17	31	33				
P10	2D2					34				
P11	2D3					35				
P12	2D4					36				
P13	2D5					37				
P14	2D6					38				
P15	2D7					39				
P16	2D8					40				

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TABLE XVI. Register operation (input minimum setup time check) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
P17	1S	Register operation (input minimum setup time check)	22,6,5,4,3,11,9,8,7	Open	2	21	23	800	1200	ns	13,15
P18	1S						24				
P19	1S						25				
P20	1S						26				
P21	1S						27				
P22	1S						28				
P23	1S						29				
P24	1S						30				
P25	2S		32,16,15,14,13,20,19,18,17		12	31	33				
P26	2S						34				
P27	2S						35				
P28	2S						36				
P29	2S						37				
P30	2S						38				
P31	2S						39				
P32	2S						40				
P33	1P/S		6	2	22	21	23				
P34	2P/S		16	12	32	31	33				

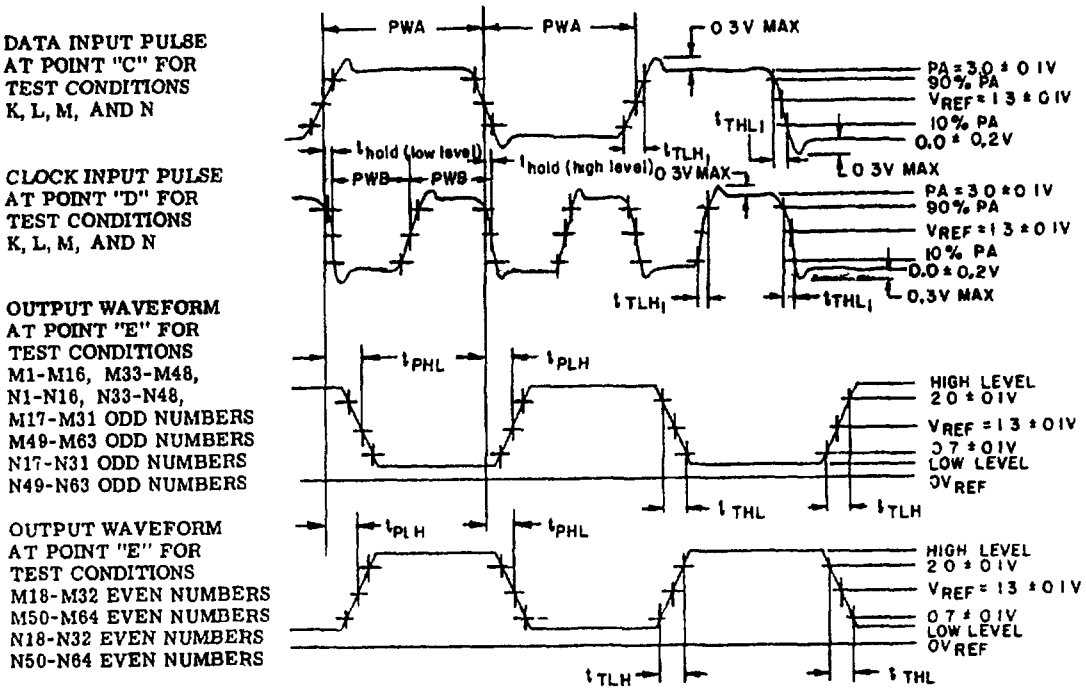
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NOTES
1. R1 = R2 = 51Ω ±5%, 1 W
2 Load all outputs of circuit under test as specified on figure 9

FIGURE 13. Transition time (t_{TLH} and t_{THL}), propagation delay time (t_{PLH} and t_{PHL}), and register operation (input minimum setup time check) test circuit for table VII, VIII, XIV, XV, and XVI.

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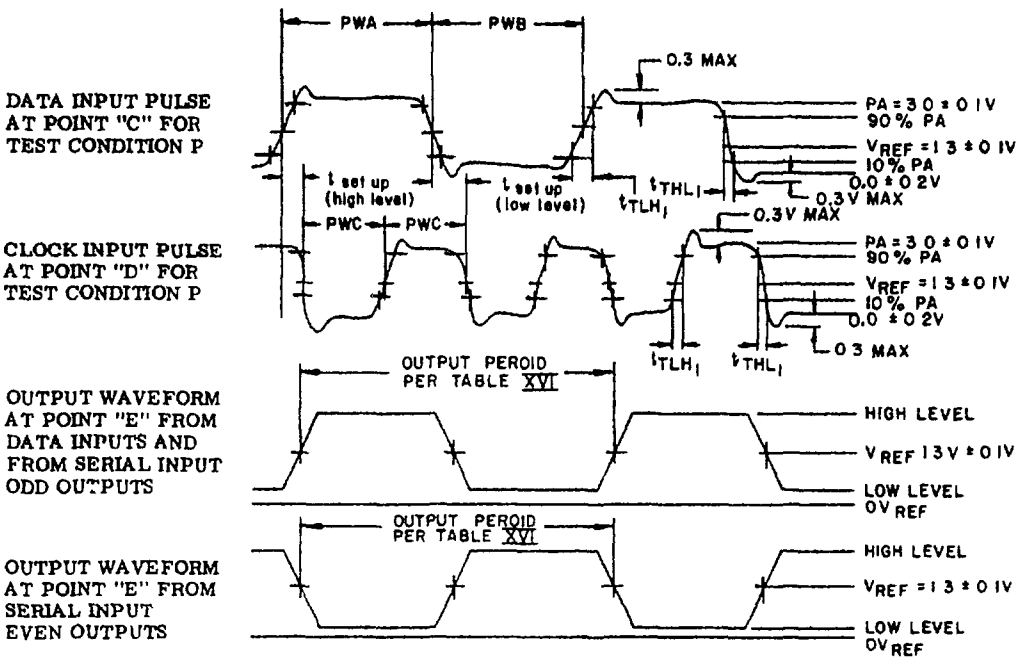


NOTES

1. Point C input pulse parameters
   a. PWA = 400 ±40 ns
   b. $t_{THL_1} = t_{TLH_1}$ = 10 ±2 ns.
2. Point D input pulse parameters·
   a. PWB = $200 \begin{smallmatrix}+10\\-0\end{smallmatrix}$ ns
   b. $t_{THL_1} = t_{TLH_1}$ = 10 ±2 ns.
3. t_{hold} (high level) = t_{hold} (low level) = $0 \begin{smallmatrix}+2\\-0\end{smallmatrix}$ ns
4. Test conditions K, L, M and N verifies minimum data hold time and minimum clock pulse width
5. All voltage levels are referenced to ground
6. For test conditions M17-M32, M49-M64, N17-N32 and N49-N64 allow a minimum of 8 clock pulses before making measurement.
7. Before making the measurement, verify that the output pulse high level is ≥ 2.4 V and the low level is ≤ 0.3 V.

FIGURE 14 Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XII, XIII, XIV, and XV

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NOTES:

1. Input parameters for test conditions P1-P32.
   a. PWA = 480 ±20 ns.
   b. $t_{THL_1} = t_{TLH_1} = 10 \pm 2$ ns.
   c. PWB = 520 ±20 ns.
   d. PWC = 250 ±10 ns
   e. t_{setup} (high level) = $100 \, ^{+5}_{-0}$ ns
   f. t_{setup} (low level) = $120 \, ^{+5}_{-0}$ ns.
2. Input parameters for test conditions P33 and P34.
   a. PWA = 475 ±20 ns.
   b. $t_{THL_1} = t_{TLH_1} = 10 \pm 2$ ns.
   c. PWB = 525 ±20 ns.
   d. PWC = 250 ±10 ns.
   e. t_{setup} (high level) = $200 \, ^{+10}_{-0}$ ns.
   f. t_{setup} (low level) = $225 \, ^{+10}_{-0}$ ns.
3. Test conditions P1-P32 verify minimum data setup time. Test conditions P33 and P34 verify minimum P S setup time
4. All voltage levels are referenced to ground

FIGURE 15 Register operation (input minimum setup time check) test circuit waveforms for table XVI.

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TABLE XVII. Output crosstalk (low-level and high-level) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q1	1Q1 Output crosstalk, low level	6	Open	5,4,3,11,9,8,7,16,15,14, 13,20,19,18,17	23	0	400	mV	16,17
Q2	1Q2	5		6,4,3,11,9,8,7,16,15,14, 13,20,19,18,17	24				
Q3	1Q3	4		6,5,3,11,9,8,7,16,15,14, 13,20,19,18,17	25				
Q4	1Q4	3		6,5,4,11,9,8,7,16,15,14, 13,20,19,18,17	26				
Q5	1Q5	11		6,5,4,3,9,8,7,16,15,14, 13,20,19,18,17	27				
Q6	1Q6	9		6,5,4,3,11,8,7,16,15,14, 13,20,19,18,17	28				
Q7	1Q7	8		6,5,4,3,11,9,7,16,15,14, 13,20,19,18,17	29				
Q8	1Q8	7		6,5,4,3,11,9,8,16,15,14, 13,20,19,18,17	30				
Q9	2Q1	16		6,5,4,3,11,9,8,7,15,14, 13,20,19,18,17	33				
Q10	2Q2	15		6,5,4,3,11,9,8,7,16,14, 13,20,19,18,17	34				
Q11	2Q3	14		6,5,4,3,11,9,8,7,16,15, 13,20,19,18,17	35				
Q12	2Q4	13		6,5,4,3,11,9,8,7,16,15, 14,20,19,18,17	36				
Q13	2Q5	20		6,5,4,3,11,9,8,7,16,15, 14,13,19,18,17	37				
Q14	2Q6	19		6,5,4,3,11,9,8,7,16,15, 14,13,20,18,17	38				

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TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

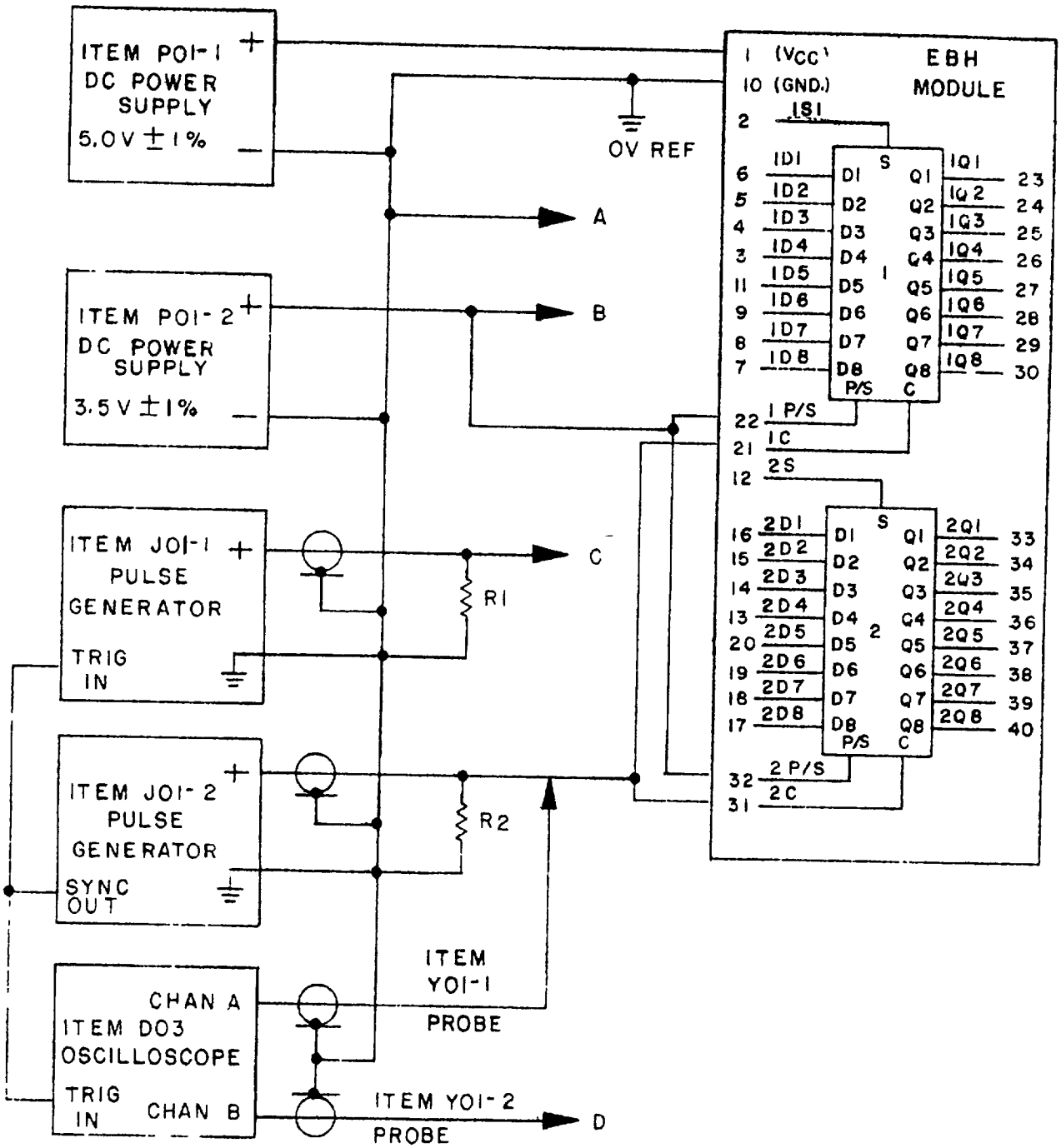
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q15	2Q7 Output crosstalk, low level	18	Open	6,5,4,3,11,9,8,7,16,15, 14,13,20,19,17	39	0	400	mV	16,17
Q16	2Q8 Output crosstalk, low level	17		6,5,4,3,11,9,8,7,16,15, 14,13,20,19,18	40				
R1	1Q1 Output crosstalk, high level	Open	6	5,4,3,11,9,8,7,16,15,14, 13,20,19,18,17	23				
R2	1Q2		5	6,4,3,11,9,8,7,16,15,14, 13,20,19,18,17	24				
R3	1Q3		4	6,5,3,11,9,8,7,16,15,14, 13,20,19,18,17	25				
R4	1Q4		3	6,5,4,11,9,8,7,16,15,14, 13,20,19,18,17	26				
R5	1Q5		11	6,5,4,3,9,8,7,16,15,14, 13,20,19,18,17	27				
R6	1Q6		9	6,5,4,3,11,8,7,16,15,14, 13,20,19,18,17	28				
R7	1Q7		8	6,5,4,3,11,9,7,16,15,14, 13,20,19,18,17	29				
R8	1Q8		7	6,5,4,3,11,9,8,16,15,14, 13,20,19,18,17	30				
R9	2Q1		16	6,5,4,3,11,9,8,7,15,14, 13,20,19,18,17	33				
R10	2Q2		15	6,5,4,3,11,9,8,7,16,14, 13,20,19,18,17	34				
R11	2Q3		14	6,5,4,3,11,9,8,7,16,15, 13,20,19,18,17	35				

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TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R12	2Q4 Output crosstalk, high level	Open	13	6,5,4,3,11,9,8,7,16,15, 14,20,19,18,17	36	0	400	mV	16,17
R13	2Q5		20	6,5,4,3,11,9,8,7,16,15, 14,13,19,18,17	37				
R14	2Q6		19	6,5,4,3,11,9,8,7,16,15, 14,13,20,18,17	38				
R15	2Q7		18	6,5,4,3,11,9,8,7,16,15, 14,13,20,19,17	39				
R16	2Q8		17	6,5,4,3,11,9,8,7,16,15, 14,13,20,19,18	40				

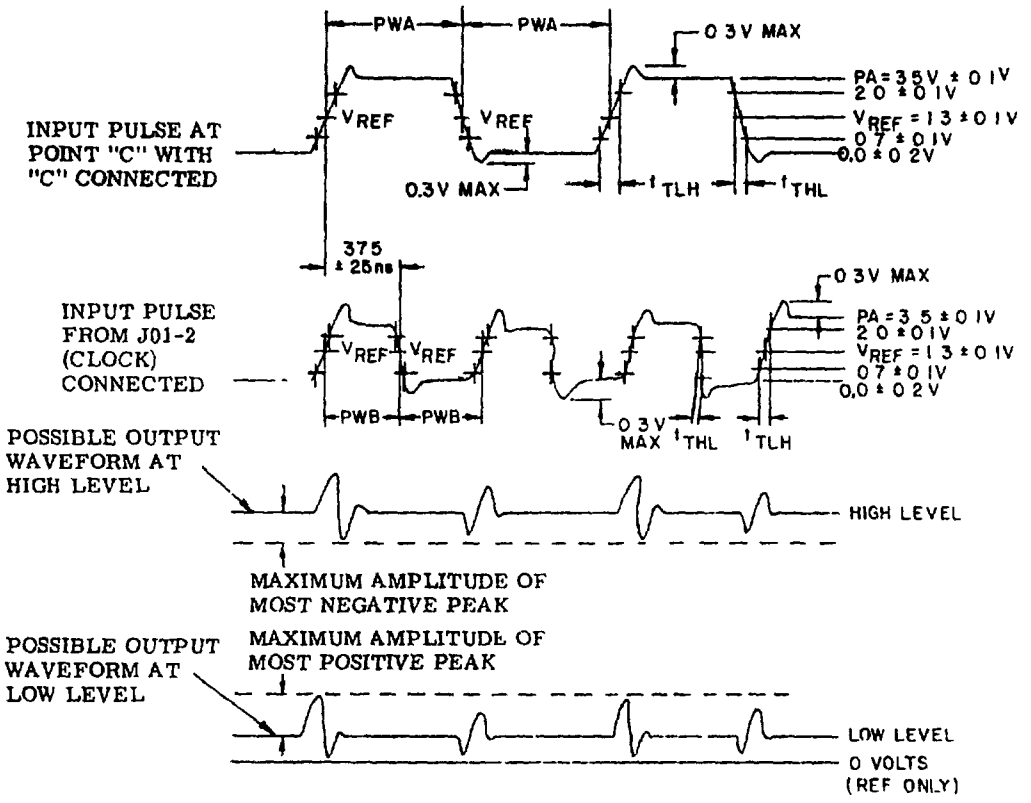
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NOTES
1. R1 = R2 = 51Ω ±5%, 1 W
2 Load all outputs with the load specified on figure 9

FIGURE 16. Output crosstalk (low-level and high-level) test circuit for table XVII

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NOTES: Input pulse parameters
PWA = 500 ±50 ns.
PWB = 250 ±25 ns.
$t_{THL} = t_{TLH}$ = 3.5 ±1.5 ns (verified at all inputs).

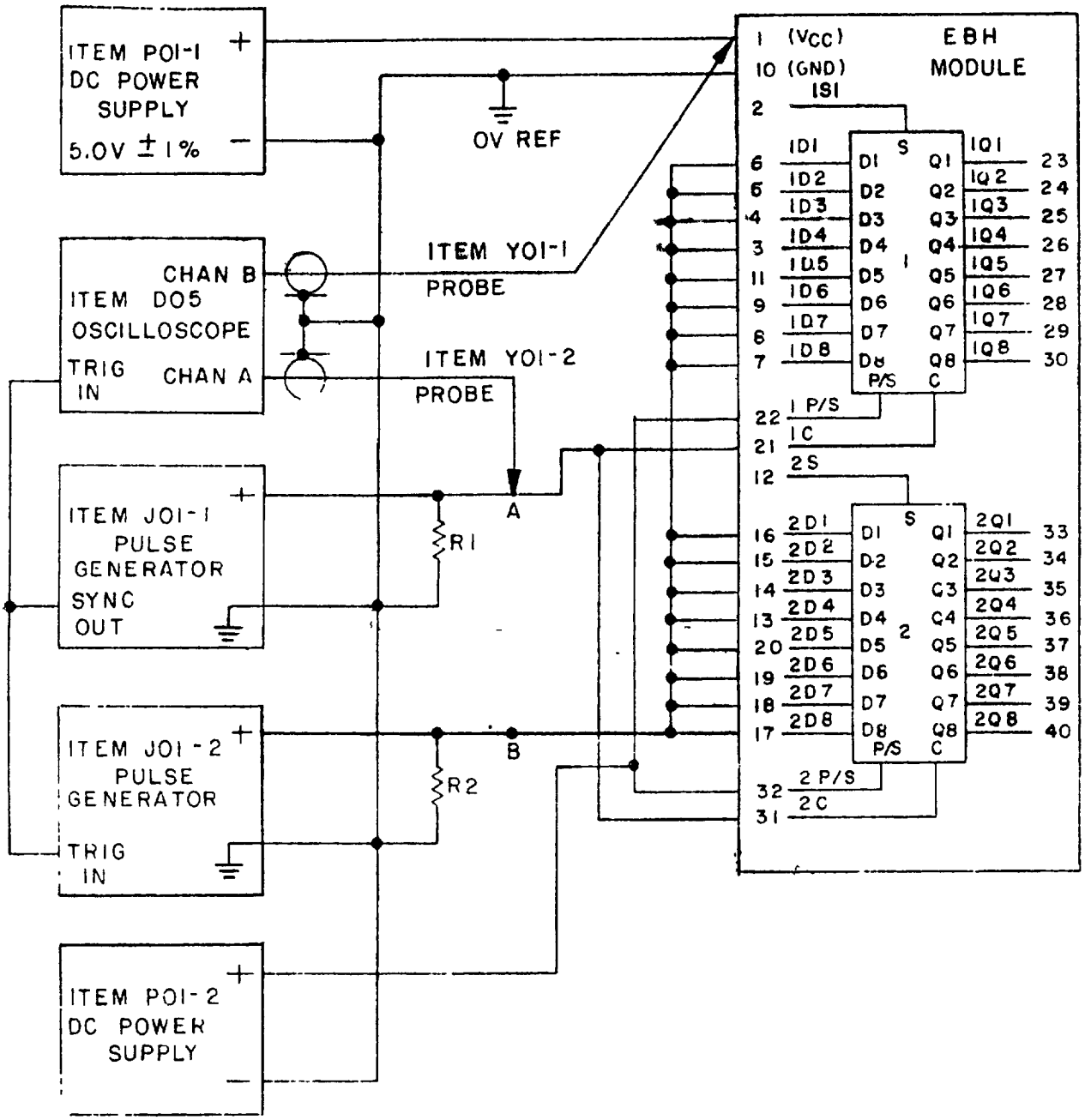
FIGURE 17. Output crosstalk (low-level and high-level) test circuit waveforms for table XVII.

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TABLE XVIII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
S	Power supply transient amplitude	0	300	mV	18,19

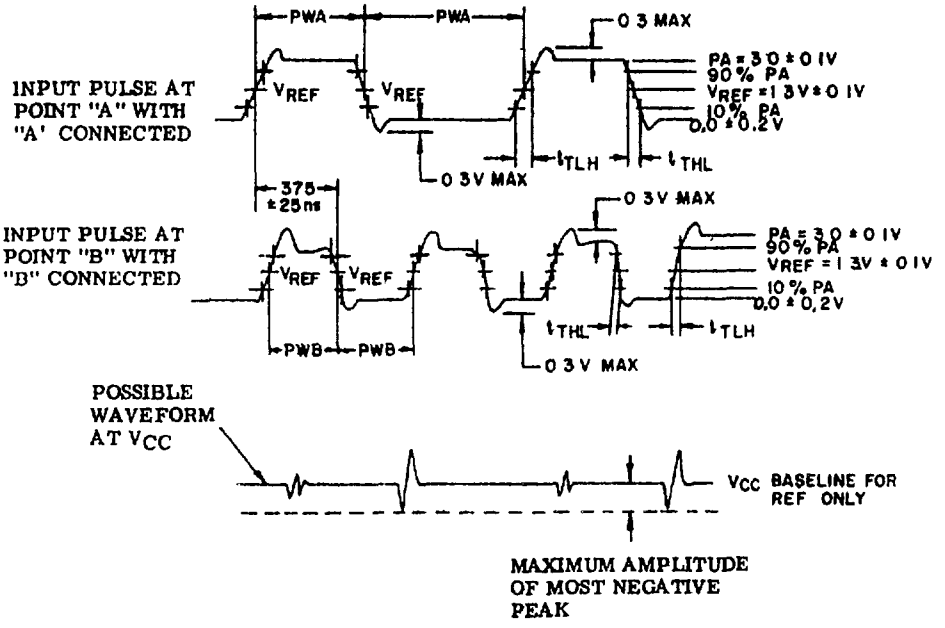
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NOTE R1 - R2 = 51Ω ±5%, 1 W

FIGURE 18 Power supply transient amplitude test circuit for table XVIII.

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NOTES: Input pulse parameters
PWA = 500 ±50 ns
PWB = 250 ±50 ns.
$t_{TLH} = t_{THL} = 10 \pm 2$ ns.

FIGURE 19. Power supply transient amplitude test circuit waveforms for table XVIII.

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TABLE XIX. Power supply filter capacitance test.

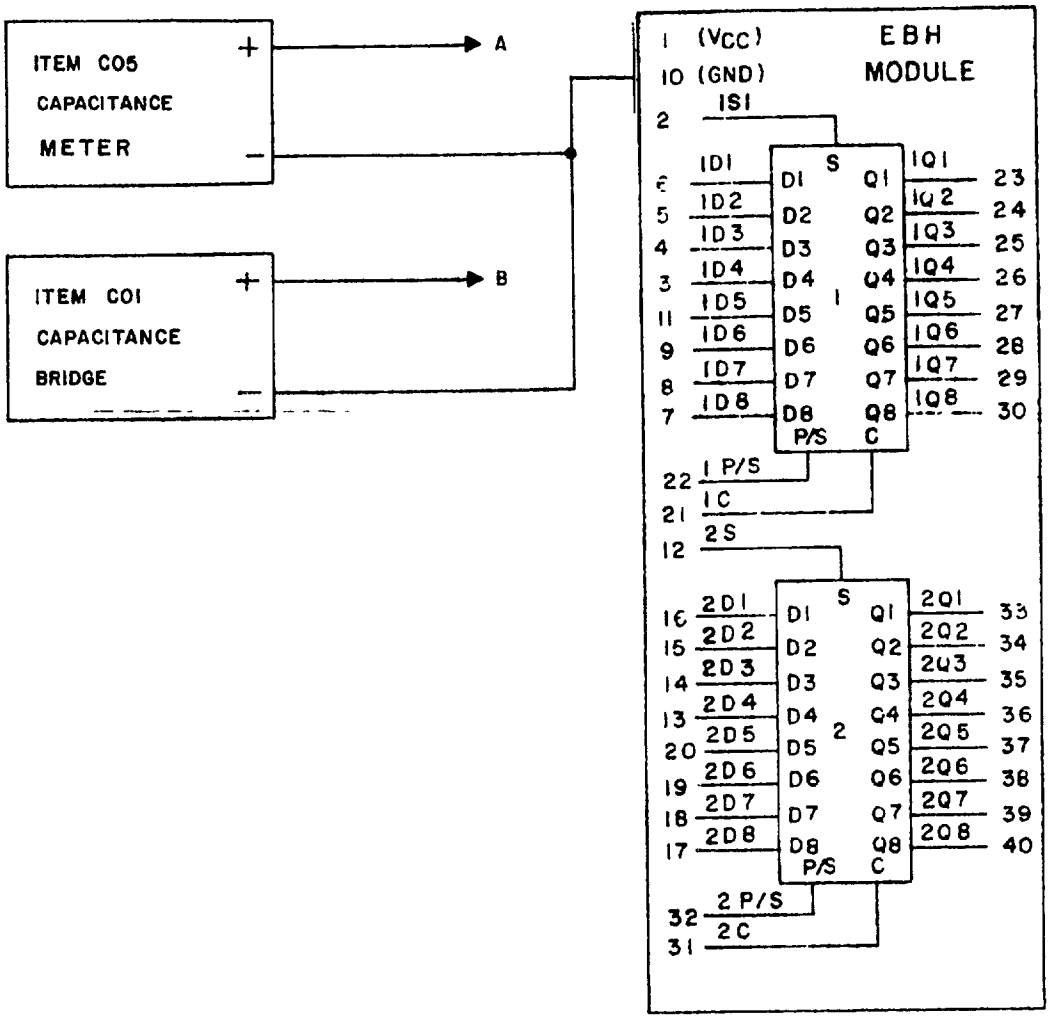
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
T	Power supply filter capacitance	Open	1	1.65	2.75	µF	20

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TABLE XX. Input capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
U1	1S Input capacitance	2	Open	2	15	pF	20
U2	1D1	6					
U3	1D2	5					
U4	1D3	4					
U5	1D4	3					
U6	1D5	11					
U7	1D6	9					
U8	1D7	8					
U9	1D8	7					
U10	2S2	12					
U11	2D1	16					
U12	2D2	15					
U13	2D3	14					
U14	2D4	13					
U15	2D5	20					
U16	2D6	19					
U17	2D7	18					
U18	2D8	17					
U19	1P/S	22			30		
U20	1C	21					
U21	2P/S	32					
U22	2C	31					

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NOTES.

1. Null out the effect of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (C01) setup:
   Frequency = 120 Hz
   V_{pk} = 0.3 V maximum.

FIGURE 20 Power supply filter capacitance and input capacitance test circuit for tables XIX and XX.

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